

Virgo General Dataset Creation Summary

Notebook

virgo-general-dataset-creation(1).ipynb

Phase

Instruction Dataset Preparation

Purpose

Creates the Virgo General instruction-following dataset by collecting, cleaning, merging, and formatting conversational datasets for supervised fine-tuning.

Objective

Build a high-quality conversational dataset suitable for Virgo General by combining multiple public datasets with custom identity and dialogue samples.

Primary Data Sources

OpenAssistant (OASST1), UltraChat 200K, OpenHermes 2.5, Virgo Identity Dataset, custom greeting samples, emotional support samples, and small-talk conversations.

Dataset Processing Pipeline

Load datasets → Extract user-assistant pairs → Remove duplicates → Filter English conversations → Remove deleted or low-quality samples → Clean URLs, emails, phone numbers and HTML → Normalize whitespace → Shuffle dataset → Merge all sources.

Identity Alignment

References to ChatGPT, GPT, OpenAI and similar assistant names are replaced with 'Virgo'. AI self-identification phrases are cleaned to establish a consistent Virgo identity.

Quality Filtering

Filters short prompts and responses, removes duplicate conversations, excludes responses containing URLs, retains English-only conversations, and enforces minimum response length.

Additional Custom Data

Adds handcrafted identity conversations, greetings, emotional support responses and small-talk examples to improve conversational ability and model personality.

Conversation Format

Conversations are formatted using BOS/EOS tokens with structured Prompt/Response sections before tokenization.

Dataset Statistics

Computes dataset size, average prompt length, average response length and token-length distribution for quality verification.

Output

Exports the cleaned dataset as `virgo_chat_dataset.csv`, ready for the tokenization stage.

Engineering Highlights

Multi-source dataset integration, deterministic shuffling (`random_state=42`), extensive text cleaning, identity alignment, automated quality filtering and scalable preprocessing pipeline.

Summary

This notebook constructs the complete Virgo General supervised fine-tuning dataset by integrating multiple high-quality conversational corpora into a unified, cleaned and standardized dataset.